

Pay More or Search Wider after Universal Rejection?

A Complete Theory Note for a Spatial Rescue Mechanism

28 August 2026

Abstract

This note develops and audits an order-level rescue mechanism for a representative ride request. The platform commits to an anonymous transaction-price path (p_1, p_2) and, when allowed, a rescue-footprint multiplier s . First-window incumbents have already seen the focal order and may reject strategically in anticipation of the same order’s rescue terms. Universal rejection therefore screens incumbent costs, becomes public information for the rider, and triggers competition among retained rejectors, time-homogeneous fresh arrivals, and newly reached outer drivers. Every policy is evaluated at its induced cutoff weak perfect Bayesian equilibrium (WPBE) before the platform optimizes.

Under the maintained uniform-Poisson, type-independent-retention, anonymous- lottery model, each feasible policy has a unique incumbent cutoff and a unique completion outcome, up to payoff-equivalent action labels. We derive an exact rider aggregation, prove that the incremental values of fixed-footprint replenishment and expanded reach vanish in both the thin and thick limits, and show that any strictly positive gap must attain an interior maximum. A closed- form ordering of the two peaks is valid in a frozen-hazard Poisson benchmark, but it is not a general theorem of the full WPBE model; a maintained-primitives fixed-menu counterexample reverses the order. Finally, a failure-contingent committed-reach capacity cost yields a finite global no-search threshold under a uniform regularity condition and a complete-WPBE local search index. An executed-notification cost does not generally have this property at a rider rescue-activation boundary.

The note also records the exact contribution boundary after an adversarial literature search through 28 August 2026. Two-round radius adjustment, joint pricing and spatial matching, strategic driver rejection, and rejection-aware notifications each exist in prior work. The defensible object is their interaction at a focal order’s public universal-rejection node.

1 Research object and audited contribution boundary

The model studies a constrained posted mechanism, not an unrestricted direct mechanism. Before the order is posted, the platform commits to an anonymous menu that may depend on the public event of universal first-window rejection, but not on realized driver counts, retention realizations, or private costs. The central question is when recovery should rely on a higher common second-window transaction price, a wider notification footprint, or both.

The closest literatures meet at the post-failure node but abstract from different sides of it. Hu, Hu, and Zhu [1] provide the closest temporal-pricing benchmark: they study a two-period price path, strategic temporal responses, and incoming drivers, but explicitly remove matched-driver rejection from their main mechanism. Qin, Yang, and Liu [2] already propose adaptive two-round broadcasting radii, so expanded radius after first-round nonresponse is not new. Ekbatalani et al. [3, 4] optimize non-exclusive notification sets under rejection uncertainty and study long-run thickness, while explicitly holding driver acceptance behavior fixed under the notification intervention. The new

working paper by Wang, Zhang, Feng, and Ke [5] studies pricing and matching under broadcasting with endogenous distance-sensitive driver response. It eliminates any broad claim that joint price and broadcast-footprint design is new. Its full text remains the highest-priority manual overlap check.

Table 1: Adversarial closest-paper map.

Paper	Central overlap	Main missing element relative to this note
Hu et al. (2022)	Two-period price, strategic temporal response, incoming drivers	No focal-order rejection screening or rescue footprint
Qin et al. (2025)	Two-round broadcast radii and heterogeneous acceptance	No announced $p_1 \rightarrow p_2$ same-order holdout equilibrium in the public record
Ekbatani et al. (2026)	Rejection-aware notifications, fresh availability, and long-run thickness	Acceptance response is stochastic/exogenous, not a price-induced WPBE
Wang et al. (2026)	Broadcasting with endogenous distance response; pricing and matching	Public abstract is one-shot/steady-state, not failure-contingent same-order rescue
Chen and Hu (2020)	Forward-looking two-sided pricing with Poisson arrivals	Market-level waiting, not rejection of a focal order
Castro et al. (2022)	Strategic decline and request reoffer	Priority/reoffer mechanism, not a common p_2 plus spatial scope
Xu and Yan (2026)	Joint state-dependent spatial dispatch and pricing	Binding dispatch/MDP, not nonbinding broadcast and holdout
This note	Same-order holdout, fresh supply, rider failure decision, and reach	Cutoff-WPBE nested inside posted (p_1, p_2, s) design

The resulting contribution claim is intentionally joint rather than ingredient-by-ingredient:

We study an order-level recovery mechanism in which public universal rejection simultaneously screens incumbent private costs, governs rider continuation, and triggers a common second-window transaction price that pools retained rejectors with fixed-footprint fresh arrivals and, when chosen, a newly reached outer annulus. The platform designs (p_1, p_2, s) subject to the induced cutoff-WPBE.

We do *not* claim the first two-period ride-hailing price, the first strategic driver rejection, the first two-round broadcast, the first matching- radius decision, or the first joint pricing-and-spatial-matching problem.

2 Environment

2.1 Prices, types, and local thickness

There is one representative order and two equal-length response windows. Rider value and drivers' persistent order costs are independent and uniform:

$$v, c \sim U[0, 1].$$

The rider and incumbent delay factors are $\beta, \delta \in (0, 1]$.

The variable p_j has one economic meaning throughout: it is the anonymous full-pass-through transaction price in window j . The rider pays p_j and the assigned driver receives p_j . Thus the current two-price model is neither a platform-profit model nor a separate fare-and-wage model. Separating the rider fare and driver wage would enlarge the mechanism to at least four monetary instruments. The platform's objective below values completion and charges a search-resource cost.

The first-window incumbent cohort and second-window fresh core cohort are independent:

$$N_1 \sim \text{Pois}(m), \quad N_2^C \sim \text{Pois}(m), \quad N_1 \perp N_2^C.$$

Here m is a *local per-window driver intensity* faced by a representative order. Because simultaneous orders and overlapping notifications are not modeled, m should not be interpreted as system-wide market thickness without a separate sparse/mean-field foundation.

2.2 Supply architectures

We compare three supply environments.

- I. **Retained incumbents only** (I): the terminal pool contains first-window rejectors who remain available. The platform chooses (p_1, p_2) .
- II. **Fixed footprint with fresh arrivals** (A): the same retained rejectors compete with an independent, time-homogeneous fresh core cohort in the second window. The platform chooses (p_1, p_2) .
- III. **Expanded rescue search** (E): the fixed-footprint fresh cohort is present under both repeat and rescue, and rescue can additionally reach an outer annulus. The platform chooses (p_1, p_2, s) .

Architectures I and A are supply counterfactuals, not nested mechanism sets. Architecture A is nested in E because $s = 1$ exactly reproduces the fixed footprint.

Let $\omega \in [0, 1]$ be the probability that a first-window rejector remains physically available and eligible for the focal order's terminal lottery. Retention is independent of cost and of the rider's terminal action. The same ω enters the aggregate pool and the focal incumbent's continuation payoff.

2.3 Search multiplier and winner-paid pickup cost

Normalize the core radius to R_0 . The search control $s \in [1, \bar{s}]$ is an *area multiplier*:

$$R(s) = R_0 \sqrt{s}.$$

Spatial Poisson independent increments therefore give an outer fresh cohort

$$N_2^O(s) \sim \text{Pois}(m(s-1)), \quad N_2^O(s) \perp N_2^C.$$

Let $u = (\ell/R_0)^2$ denote area rank. A selected outer driver pays the extra pickup cost

$$d(u) = \tau(\sqrt{u} - 1)_+, \quad \tau \geq 0,$$

after assignment. Core pickup cost is absorbed into c . This timing is a maintained assumption: an unassigned notified driver pays no sunk relocation cost. If pickup or activation cost were paid before assignment, fresh entry would require a congestion fixed point rather than the Poisson thinning below.

2.4 Timing and information

1. The platform publicly commits to the feasible anonymous menu.
2. A rider observes v and posts iff $v \geq p_1$.
3. The N_1 incumbents receive the focal order. Each observes the order, (p_1, p_2, s) , and own c , but not other costs or realized driver counts.
4. Incumbents simultaneously accept at p_1 or reject and wait. If at least one accepts, the order is assigned by an anonymous lottery.
5. Universal rejection is publicly observed. The rider does not observe the number of first-window drivers, retention, future arrivals, or volunteers. She chooses abandon, repeat, or rescue using equilibrium beliefs.
6. Only after the terminal action is selected do fresh drivers first see the order. They volunteer when the assigned-order surplus is nonnegative. Retained incumbents and fresh volunteers enter a common anonymous lottery.

The platform may condition on public universal rejection but not on hidden or realized counts. The reach commitment must be verifiable or technologically enforceable.

3 Conditional supply and assignment

For a generic cost distribution F , let a be a proposed first-window incumbent cutoff. A retained incumbent can volunteer at terminal price p_j only when $a < c \leq p_j$, so define

$$I_j^\omega(a) = \omega m [F(p_j) - F(a)]_+.$$

Under the maintained uniform distribution, write $F(x) = [x]_0^1$.

Define the fresh willing-driver intensities

$$e_j^I = 0, \quad e_j^A = mF(p_j),$$

and, in the expanded architecture,

$$e_1^E = mF(p_1), \quad e_2^E = mF(p_2) + m\sigma_{p_2}(s),$$

where

$$\sigma_p(s) = \int_1^s F(p - d(u)) du.$$

The total terminal willing-driver intensity and coverage are

$$\Lambda_j^r(a) = I_j^\omega(a) + e_j^r, \quad C_j^r(a) = 1 - e^{-\Lambda_j^r(a)}.$$

Lemma 1 (Conditional Poisson supply decomposition). *Under a first-window cutoff a , conditioning on universal rejection leaves the number of terminal-eligible rejectors Poisson with mean $I_j^\omega(a)$. Independent fresh core and outer-annulus volunteers add their respective Poisson means. Hence the terminal volunteer count is Poisson with mean $\Lambda_j^r(a)$ in architecture r .*

Proof. Mark N_1 by driver cost. Poisson splitting makes the counts with $c \leq a$ and $a < c \leq p_j$ independent, with means $mF(a)$ and $m[F(p_j) - F(a)]$. Universal rejection sets the first count to zero but does not change the second count's conditional distribution. Independent ω -retention thins the latter to mean $I_j^\omega(a)$. The fresh core flow is independent. Spatial independent increments and cost thinning give outer mean $m\sigma_{p_2}(s)$. Sums of independent Poisson variables are Poisson with the sum of their means. $\square$

If a focal volunteer faces $N \sim \text{Pois}(z)$ opponents in an anonymous lottery, her expected assignment probability is

$$h(z) = \mathbb{E} \left[\frac{1}{N+1} \right] = \frac{1 - e^{-z}}{z}, \quad h(0) = 1.$$

An outer fresh driver's expected payoff is

$$h(\Lambda_2)[p_2 - c - d(u)].$$

Because the pickup cost is assignment-contingent, $h(\Lambda_2) > 0$ scales the payoff without changing its sign. Thus volunteering is equivalent to $c + d(u) \leq p_2$, which validates the thinning formula for $\sigma_{p_2}(s)$.

4 Rider continuation and exact aggregation

After universal rejection, a rider of value v compares

$$0, \quad C_1(a)(\beta v - p_1), \quad C_2(a)(\beta v - p_2).$$

Let $\eta_j(a)$ be the conditional mass of posted riders selecting terminal action $j \in \{0, 1, 2\}$. Ties at a single rider type are immaterial. If two terminal actions are payoff-identical for a positive measure, we use the deterministic convention abandon before repeat before rescue. This convention selects labels only; completion and incumbent waiting payoffs coincide across payoff-identical terminal actions.

Set

$$P = 1 - p_1, \quad q = e^{-ma}, \quad \rho_j = 1 - \frac{p_j}{\beta}, \quad A_j = \rho_j C_j.$$

Lemma 2 (Exact rider aggregation). *Suppose $0 \leq C_1 \leq C_2 \leq 1$. If $C_2 > C_1$, the repeat-rescue crossing is*

$$v^{12} = \frac{C_2 p_2 - C_1 p_1}{\beta(C_2 - C_1)}, \quad v^{12} - \frac{p_2}{\beta} = \frac{C_1(p_2 - p_1)}{\beta(C_2 - C_1)} \geq 0.$$

On the active domain $p_2 \leq \beta$, rescue has positive mass iff $A_2 > A_1$ and

$$P\eta_2 = \frac{[A_2 - A_1]_+}{C_2 - C_1}.$$

On the full price domain, terminal completion integrated over the original rider population is

$$P(\eta_1 C_1 + \eta_2 C_2) = \max\{0, A_1, A_2\}.$$

Consequently every equilibrium completion outcome satisfies

$$M = P(1 - q) + q \max\{0, A_1, A_2\}. \quad (1)$$

If $C_2 = C_1$ and $p_2 > p_1$, repeat dominates rescue. If both coverage and price are equal, action labels are tie-rule dependent but completion is not.

Proof. The three continuation payoffs are affine in v and their upper envelope has the order abandon, repeat, rescue. On a strict rescue branch, riders in $(v^{12}, 1]$ rescue and direct algebra gives

$$1 - v^{12} = \frac{A_2 - A_1}{C_2 - C_1}.$$

Moreover,

$$C_1 \left(v^{12} - \frac{p_1}{\beta} \right) + C_2(1 - v^{12}) = A_2.$$

If rescue is inactive, repeat contributes $[A_1]_+$; if repeat is also inactive, terminal completion is zero. Multiplying by the first-window failure probability yields (1). $\square$

For later reference, expected executed outer notifications on a strict rescue branch are

$$Q^{\text{ex}} = Pq\eta_2 m(s - 1) = qm(s - 1) \frac{[A_2 - A_1]_+}{C_2 - C_1}. \quad (2)$$

The ratio in (2) is not defined by continuity at a joint price-coverage degeneracy; this is one reason not to use executed contacts as the main cost primitive.

5 Cutoff-WPBE

Under a proposed aggregate cutoff a , a type- c incumbent's accept-minus- wait payoff in architecture r is

$$D_r(c; a) = h(ma)(p_1 - c) - \delta\omega e^{-ma} \sum_{j=1}^2 \eta_j^r(a) h(\Lambda_j^r(a)) [p_j - c]_+. \quad (3)$$

The first term incorporates first-window assignment competition. The second requires universal rejection by other incumbents, survival, rider continuation, and terminal assignment.

Definition 3 (Cutoff-WPBE). A symmetric cutoff $a \in [0, p_1]$, together with rider and fresh-driver strategies, is a cutoff-WPBE if: (i) incumbents accept for $c < a$ and reject for $c > a$, with the appropriate weak inequalities at a ; (ii) riders and fresh drivers are sequentially rational; and (iii) on-path beliefs after universal rejection are generated by the cutoff and Lemma 1.

Theorem 4 (Unique incumbent cutoff and completion outcome). *Fix a feasible policy in I, A , or E . Under uniform driver costs, type-independent retention, cutoff-independent fresh intensities, and the anonymous lottery, there is exactly one incumbent cutoff $a^* \in [0, p_1]$. The cutoff and completion outcome are continuous in policy and primitives when fresh intensities are continuous. Full strategies are unique only up to payoff-equivalent rider labels and measure-zero driver ties.*

The proof is in Appendix A. The key inequality is that at every interior cutoff root, the immediate-acceptance residual crosses zero strictly downward. The restrictions in the theorem are substantive. Type-dependent retention, pre-assignment entry costs, priority assignment, or general F may destroy the derivative ordering and require a new equilibrium analysis.

Corollary 5 (Flat price). *For $p_1 = p_2 = p$ and $s = 1$, the unique incumbent cutoff is $a = p$ in every architecture.*

Proof. For $a < p$, terminal service has the same price and weakly worse timing and assignment odds than immediate acceptance. Formally, the residual factors as $(p - a)$ times a strictly positive bracket because $h(ma) \geq e^{-ma}$, $\delta\omega \leq 1$, and terminal competition cannot improve the focal driver's assignment probability. Thus no $a < p$ is a root; at $a = p$, all lower types accept and higher types have no positive surplus. $\square$

6 Outer posted-mechanism design

The feasible policy sets are compact:

$$\begin{aligned}\mathcal{M}_I &= \mathcal{M}_A = \{(p_1, p_2) : 0 \leq p_1 \leq p_2 \leq 1\}, \\ \mathcal{M}_E &= \{(p_1, p_2, s) : 0 \leq p_1 \leq p_2 \leq 1, 1 \leq s \leq \bar{s}\}.\end{aligned}$$

Policies with active delayed service may impose $p_2 \leq \beta$ without loss; the inactive closure is retained for existence.

Free, unbounded reach would make completion weakly favor the largest feasible scope. We therefore distinguish the driver's winner-paid pickup cost from a platform search resource. The main specification prices the capacity needed to honor the promised outer footprint after first-window failure:

$$Q^{\text{com}} = (1 - p_1)e^{-ma}m(s - 1). \quad (4)$$

This cost is failure-contingent but does not vanish merely because the rider is exactly indifferent at the rescue-activation boundary. It can represent reserved notification bandwidth, dispatch attention, or the shadow cost of shared capacity omitted from the representative-order partial equilibrium. Core notification infrastructure is normalized to zero incremental cost.

For completion value $B > 0$ and per-unit capacity cost $\kappa \geq 0$, define

$$J_r^{\text{com}}(\mu, a) = BM_r(\mu, a) - \kappa Q_r^{\text{com}}(\mu, a), \quad Q_I^{\text{com}} = Q_A^{\text{com}} = 0.$$

The equilibrium-constrained value is

$$\mathcal{V}_r(m; \kappa) = \max_{\mu \in \mathcal{M}_r} \min_{a \in \mathcal{E}_r^{\text{WPBE}}(\mu; m)} J_r^{\text{com}}(\mu, a). \quad (5)$$

Theorem 4 makes the inner set a singleton, but writing the problem this way makes clear that the platform does not optimize at a fixed cutoff. Computationally, one may fix p_1 , optimize p_2 and s while re-solving the WPBE at every evaluation, and then optimize p_1 .

The objective is completion net of a search-resource cost. It is not platform profit or total welfare. A profit model would separate rider fare and driver wage and would also include pickup-time cancellation or other service costs.

7 What market thickness guarantees

Let $V_I(m)$ and $V_A(m)$ denote optimized completion in architectures I and A . Define

$$\Delta_A(m) = B[V_A(m) - V_I(m)], \quad \Delta_E(m; \kappa) = \mathcal{V}_E(m; \kappa) - BV_A(m).$$

The first difference compares two supply environments and need not be nonnegative by set inclusion. The second is nonnegative because $s = 1$ reproduces architecture A at zero outer-capacity cost.

Theorem 6 (Endpoint collapse under WPBE and full reoptimization). *For finite $\bar{s}$ and compact policy sets,*

$$\lim_{m \downarrow 0} V_I(m) = \lim_{m \downarrow 0} V_A(m) = 0, \quad \lim_{m \downarrow 0} \mathcal{V}_E(m; \kappa) = 0,$$

and

$$\lim_{m \rightarrow \infty} V_I(m) = \lim_{m \rightarrow \infty} V_A(m) = 1, \quad \lim_{m \rightarrow \infty} \mathcal{V}_E(m; \kappa) = B.$$

Consequently both incremental gaps vanish as $m \downarrow 0$ and $m \rightarrow \infty$. For all sufficiently large m ,

$$0 \leq \Delta_E(m; \kappa) \leq B[1 - V_A(m)] \leq B \left(\frac{\log m}{m} + \frac{1}{m} - \frac{\log m}{m^2} \right).$$

If $\kappa > 0$, any expanded-search optimizer satisfies

$$Q_m^{\text{com},*} \leq \frac{B[1 - V_A(m)]}{\kappa} = O\left(\frac{\log m}{\kappa m}\right).$$

Proof. Completion requires at least one available driver in a reached cohort whose total mean is no more than $(1 + \bar{s})m$. Hence completion is at most $1 - e^{-(1+\bar{s})m}$, and the optimized nonnegative objective vanishes in the thin limit.

For $m > e$, choose the feasible flat policy

$$p_1 = p_2 = p_m = \frac{\log m}{m}, \quad s = 1.$$

Corollary 5 gives $a = p_m$. First-window completion alone is

$$\left(1 - \frac{\log m}{m}\right) \left(1 - \frac{1}{m}\right) \rightarrow 1,$$

and this policy uses no outer capacity. Completion is at most one, proving the thick limits. Subtraction yields endpoint collapse. Nesting gives $\Delta_E \geq 0$, and the displayed flat-policy lower bound yields the rate. Optimality relative to $s = 1$ implies $BM_E - \kappa Q_m^{\text{com},*} \geq BV_A$; using $M_E \leq 1$ gives the resource bound. $\square$

Corollary 7 (Existence, not uniqueness, of an interior peak). *If $\Delta_E(m_0; \kappa) > 0$ at some finite m_0 , then it has a positive global maximizer in $(0, \infty)$. The same statement holds for Δ_A if $\Delta_A(m_0) > 0$. Neither conclusion implies a unique peak.*

Proof. Continuity follows from Theorem 4, the deterministic tie closure, and Berge's maximum theorem. Choose thin and thick endpoints at which the gap is less than half its value at m_0 . A maximum on the intervening compact interval cannot occur at either endpoint. $\square$

This theorem gives a robust scarce-intermediate-thick support result. It does not by itself identify the shape inside that support.

8 The two-peak question

8.1 A valid frozen-hazard theorem

The clean closed forms arise only when normalized hazards and activation weights are held fixed as m changes. This is a useful Poisson benchmark, not an endogenous full-design WPBE comparative static.

Theorem 8 (Ordered peaks in a frozen-hazard Poisson benchmark). *Let $A(m) > 0$ be differentiable, nonincreasing, and log-concave, and let*

$$0 \leq \lambda_I < \lambda_A < \lambda_E$$

be constants. Define gross adjacent gains

$$\Delta_A^0(m) = A(m)e^{-\lambda_I m} \left(1 - e^{-(\lambda_A - \lambda_I)m}\right),$$

$$\Delta_E^0(m) = A(m)e^{-\lambda_A m} \left(1 - e^{-(\lambda_E - \lambda_A)m}\right).$$

If both gains vanish at the endpoints, each is strictly log-concave with a unique interior maximizer, and

$$\arg \max_m \Delta_E^0(m) < \arg \max_m \Delta_A^0(m).$$

If $A(m) = e^{-\varphi m}$ and $\varphi + \lambda_I > 0$, the maximizers are

$$m_A^* = \frac{\log[(\varphi + \lambda_A)/(\varphi + \lambda_I)]}{\lambda_A - \lambda_I},$$

$$m_E^* = \frac{\log[(\varphi + \lambda_E)/(\varphi + \lambda_A)]}{\lambda_E - \lambda_A},$$

with

$$m_E^* < \frac{1}{\varphi + \lambda_A} < m_A^*.$$

Proof. For $d > 0$, $\log(1 - e^{-dm})$ has strictly negative second derivative. Hence both gains are strictly log-concave. Put $d_A = \lambda_A - \lambda_I$ and $d_E = \lambda_E - \lambda_A$. Their log-slope difference is

$$(\log \Delta_A^0)' - (\log \Delta_E^0)' = \frac{d_A}{1 - e^{-d_A m}} - \frac{d_E}{e^{d_E m} - 1} > 0,$$

because

$$\frac{d_A}{1 - e^{-d_A m}} > \frac{1}{m} > \frac{d_E}{e^{d_E m} - 1}.$$

At the search peak, the search log slope is zero, while the arrival log slope is positive; the arrival peak lies to its right. Under exponential activation, each gain is a difference of two exponentials. Solving its first-order condition yields the two closed forms. The middle inequality follows from $\log(b/a) = \int_a^b t^{-1} dt$. $\square$

8.2 Why the full WPBE does not inherit the order automatically

In the full model, prices, cutoffs, rider action masses, and per-unit hazards typically vary with m and differ across supply architectures:

$$p_j = p_j^r(m), \quad a = a^r(m), \quad \eta_j = \eta_j^r(m), \quad \lambda_j = \lambda_j^r(m).$$

Theorem 8 therefore does not prove an ordered peak after equilibrium response and reoptimization.

Remark 9 (Maintained-primitives fixed-menu reversal). The absence of a primitive theorem is not merely abstract. At

$$(\beta, \delta, \omega, \tau, \kappa) = (.1510729, .4580650, .1790836, .8263075, .0100430)$$

and the fixed menu

$$(p_1, p_2, s) = (.0371029, .0864844, 2.4731078),$$

re-solving the architecture-specific cutoff-WPBE at every m gives

$$\begin{aligned} m_A^* &= 12.59214, & \Delta_A(m_A^*) &= .167859, \\ m_E^* &= 15.50563, & \Delta_E(m_E^*) &= .002470. \end{aligned}$$

Thus $m_E^* > m_A^*$, and the search gap changes sign and has multiple lobes. These gaps use the archived executed-contact objective. The solver-certified counterexample lies inside the maintained primitives and is sufficient to make the corresponding universal fixed-menu peak-order claim unsafe. It is not an interval proof, a symbolic characterization of when reversal occurs, or a counterexample to every alternative search-cost specification.

One can nevertheless certify an ordering for a particular calibrated model by a uniform separation condition.

Proposition 10 (Uniform separation certificate). *Let a compact set K contain the reference peaks, all full-gap maximizers, and a separator $\bar{m} \in (m_E^0, m_A^0)$. Define*

$$\begin{aligned} \Gamma_A &= \Delta_A^0(m_A^0) - \sup_{m \in K: m \leq \bar{m}} \Delta_A^0(m) > 0, \\ \Gamma_E &= \Delta_E^0(m_E^0) - \sup_{m \in K: m \geq \bar{m}} \Delta_E^0(m) > 0. \end{aligned}$$

If

$$\|\Delta_A - \Delta_A^0\|_{\infty, K} < \Gamma_A/2, \quad \|\Delta_E - \Delta_E^0\|_{\infty, K} < \Gamma_E/2,$$

then every full search-gap maximizer is below $\bar{m}$, and every full arrival-gap maximizer is above $\bar{m}$.

Proof. At m_A^0 , the full arrival gap is at least $\Delta_A^0(m_A^0) - \varepsilon_A$. Anywhere weakly below $\bar{m}$, it is at most the displayed restricted supremum plus ε_A . The former is strictly larger when $2\varepsilon_A < \Gamma_A$. The search argument is symmetric. $\square$

This proposition is a robustness certificate, not a primitive sufficient condition. It should be verified numerically rather than advertised as the main source of economic intuition.

9 Costly search and the no-expansion boundary

9.1 Global committed-reach threshold

Let $\widehat{M}(p_1, p_2, s; m)$ denote equilibrium-reduced completion and

$$M_A^*(m) = \max_{(p_1, p_2) \in \mathcal{M}_A} \widehat{M}(p_1, p_2, 1; m).$$

Define

$$\kappa_{\text{com}}^*(m) = B \sup_{\mu \in \mathcal{M}_E: Q^{\text{com}}(\mu) > 0} \frac{[\widehat{M}(\mu) - M_A^*(m)]_+}{Q^{\text{com}}(\mu)}. \quad (6)$$

Theorem 11 (Exact and finite committed-reach no-search threshold). *Use the smallest- s policy when outcomes tie. Then*

$$\mathcal{V}_E(m; \kappa) = BM_A^*(m) \iff \kappa \geq \kappa_{\text{com}}^*(m).$$

Moreover, optimal committed outer capacity is weakly decreasing in κ . If $\widehat{M}$ is uniformly Lipschitz in s on the compact policy set,

$$|\widehat{M}(p_1, p_2, s) - \widehat{M}(p_1, p_2, 1)| \leq L(s - 1),$$

then $\kappa_{\text{com}}^*(m) < \infty$ for every fixed $m > 0$.

Proof. For every positive-resource outcome,

$$B\widehat{M} - \kappa Q^{\text{com}} \leq BM_A^* \iff \kappa \geq B \frac{\widehat{M} - M_A^*}{Q^{\text{com}}}.$$

Taking the positive part and supremum proves the threshold, while $s = 1$ attains BM_A^* . For resource monotonicity, add the two optimality inequalities at costs $\kappa_1 < \kappa_2$ to obtain $(\kappa_2 - \kappa_1)(Q_1 - Q_2) \geq 0$.

For finiteness, note $M_A^*(m) > 0$. Whenever $\widehat{M} > M_A^*$, completion cannot exceed posting mass, so $P = 1 - p_1 \geq \widehat{M} > M_A^*$. Also $q = e^{-ma} \geq e^{-m}$ because $a \leq 1$. The same (p_1, p_2) at $s = 1$ is feasible in A , so

$$\widehat{M}(p_1, p_2, s) - M_A^* \leq \widehat{M}(p_1, p_2, s) - \widehat{M}(p_1, p_2, 1) \leq L(s - 1).$$

Therefore

$$\frac{B[\widehat{M} - M_A^*]}{Q^{\text{com}}} \leq \frac{BLe^m}{mM_A^*(m)} < \infty.$$

□

The uniform Lipschitz condition is a regularity assumption on the equilibrium-reduced map. Unique continuity alone is not enough to deliver a finite ratio bound.

9.2 Complete-WPBE local search index

Let (p_1^A, p_2^A) be a unique interior fixed-arrival optimum with a regular cutoff, $p_2 > p_1$, and strict rescue activity at $s = 1$. Evaluate all objects at this policy and write

$$g = P\eta_2 = \frac{A_2 - A_1}{C_2 - C_1}, \quad K = \frac{p_2}{\beta} - p_1 + (1 - \omega)\rho_2 e^{-\Lambda_2}.$$

Let $\mathcal{R}(a, s) = D_E(a; a)$ and

$$a_s = -\frac{\mathcal{R}_s}{\mathcal{R}_a}$$

be the cutoff response along the regular WPBE branch.

Theorem 12 (Local no-search index with equilibrium cutoff response). *At $s = 1$, the right derivatives are*

$$M_s = e^{-ma} m [\rho_2 e^{-\Lambda_2} p_2 + K a_s], \\ Q_s^{\text{com}} = P e^{-ma} m, \quad Q_s^{\text{ex}} = g e^{-ma} m.$$

Hence the complete-WPBE local break-even indices are

$$\kappa_{\text{loc}}^{\text{com}} = \frac{B}{P} [\rho_2 e^{-\Lambda_2} p_2 + K a_s]$$

and, for executed contacts on a strict rescue branch,

$$\kappa_{\text{loc}}^{\text{ex}} = \frac{B}{g} [\rho_2 e^{-\Lambda_2} p_2 + K a_s].$$

Expansion is locally valuable precisely when the applicable κ lies below its index. A global conclusion additionally requires a nonincreasing profile derivative or a single positive-to-negative crossing.

Proof. The implicit-function theorem gives $a_s = -\mathcal{R}_s/\mathcal{R}_a$. Differentiate the rescue branch of (1). At $s = 1$, $\Lambda_{2,s} = mp_2$ and $\Lambda_{2,a} = -\omega m$. Collecting the cutoff terms gives K and the displayed M_s . Differentiating (4) and (2) gives the resource derivatives. Because prices are optimized at the fixed-arrival interior optimum, the envelope theorem removes their first-order response when s is opened. $\square$

The cutoff term is economically important. Direct reach has two opposing effects on incumbent waiting. Greater rescue coverage can move more riders into rescue, raising waiting value, while more fresh competition lowers the incumbent's assignment probability. Thus a_s is not signed globally.

9.3 Why executed contacts are not the main cost

Suppose at fixed (a, p_1, p_2) that $p_2 > p_1$, $p_2 < \beta$, $C_2(1) > C_1$, and $A_2(1) = A_1$. Let $x = s - 1$. Then rescue activation gives

$$M(s) - M(1) = \Theta(x), \quad P\eta_2(s) = \Theta(x), \quad Q^{\text{ex}}(s) = \Theta(x^2).$$

If the profiled equilibrium completion retains a strictly positive right derivative, no finite per-executed-contact cost can implement $s = 1$. This argument does not apply when $p_1 = p_2$, where the rescue mass may already be first order. Committed reach instead has $Q^{\text{com}}(s) = \Theta(x)$ and removes this denominator failure.

10 Computational falsification and implementation status

The analytical model is accompanied by an exact Poisson solver that evaluates every candidate policy at the full type-domain cutoff conditions and then performs the outer search. The currently archived numerical tables use the earlier executed-contact objective, whereas the main theory above recommends committed reach. They support the following equilibrium and implementation claims but are not a committed-cost comparative-static audit:

- all 20 unit tests pass, including regime identities, independent core arrivals, incumbent retention, spatial thinning, full best responses, and outer nestedness;
- the formal audit contains 99 environments and 297 architecture rows; all are cutoff-certified and the maximum certified equilibrium-cutoff count is one;
- the minimum numerical E -minus- A value gap is -1.22×10^{-10} , which is optimization tolerance, and the maximum is 0.123759;
- none of the 99 archived fixed-arrival optima is close to zero rescue mass (the minimum is approximately 0.0678), so the current figures are not mechanically generated by the executed-contact activation loophole;
- the maintained-primitives fixed-menu reversal reported above was found by adversarial search and re-solves the cutoff-WPBE at every thickness.

The earlier claim of 48 primitive vectors and 1,344 environments is not supported by an archived script and result table in the current folder and is therefore not used in this note.

Two numerical boundary issues remain in the implementation rather than in the analytical equilibrium engine. An absolute rider-utility tolerance of 10^{-12} can convert a strict choice into a tie

at extremely small m , and rounded rider breakpoints can collapse the conditional support as $p_1 \uparrow 1$. These cases should be replaced by the closed-form upper- envelope formulas in Lemma 2 before the code is used for extreme- boundary comparative statics.

11 Interpretation, limitations, and theorem agenda

The model is internally coherent under its maintained assumptions, but those assumptions define the scope of every theorem.

1. **Representative-order partial equilibrium.** Independent local Poisson pools ignore capacity competition across simultaneous orders. A system-wide thick-market claim requires a sparse/mean-field extension.
2. **Anonymous lottery.** Search affects the assignment probability of retained rejectors through competition. Nearest-driver or history-priority assignment can change this discipline channel.
3. **Winner-paid pickup.** The clean outer thinning relies on the pickup cost being borne only by the selected driver. Pre-assignment relocation requires endogenous entry and congestion.
4. **Type-independent exit.** Retention ω is an availability shock independent of c and has zero outside payoff. Selection in exit may change the posterior and uniqueness proof.
5. **No rider pickup disutility.** The rider discounts only across windows through β ; longer outer pickup distance does not directly lower rider utility. Adding that channel would strengthen the case for costly or interior reach.
6. **Pass-through price.** The current objective is not profit. A profit paper must separate fare and wage or justify the direct-transfer institution.

The next genuinely nontrivial theorem is not another endpoint argument. It is a primitive region in which the fully profiled scope derivative single-crosses or in which the separation margins can be verified analytically. Such a result could deliver a scarce-intermediate-thick search window after full WPBE response and reoptimization. Until then, the correct hierarchy is:

1. unique incumbent cutoff and exact completion aggregation: proved;
2. endpoint collapse and existence of a positive interior maximizer: proved;
3. finite committed-reach no-search threshold under uniform regularity: proved;
4. ordered peaks with frozen hazards: proved as a benchmark;
5. ordered peaks after full WPBE reoptimization: numerical conjecture, not a theorem;
6. global unique peaks and monotone $s^*(m)$: false or unproved.

12 Conclusion

Universal rejection is the organizing economic event. It screens the incumbent pool, informs the rider, and activates a recovery market in which one common transaction price changes both retained-driver eligibility and fresh-driver willingness. Fixed-footprint arrivals and geographic expansion restore supply through different channels: the former changes the supply environment in every terminal action, while the latter changes who first sees rescue and adds pickup friction. The platform's menu must therefore be solved as an equilibrium-constrained posted mechanism.

The strongest currently valid theoretical result is a chain rather than a single closed form: conditional Poisson supply, unique cutoff-WPBE outcome, exact rider aggregation, full-design endpoint collapse, and a committed-reach cost threshold. The elegant closed-form peak order identifies an important Poisson force but is not universal once equilibrium composition and policy reoptimization are allowed. That boundary is a result of the adversarial audit, not a defect to conceal.

A Proof of the unique-cutoff theorem

For uniform costs and $a \leq p_1 \leq p_j$,

$$\Lambda_j(a) = \omega m(p_j - a) + e_j, \quad \Lambda_{j,a} = -\omega m.$$

In all three architectures, $e_2 \geq e_1$, so

$$\Lambda_2(a) - \Lambda_1(a) = \omega m(p_2 - p_1) + (e_2 - e_1) \geq 0$$

is independent of a .

Step 1: type single crossing and existence. Away from the kinks p_1, p_2 ,

$$\frac{\partial D_r(c; a)}{\partial c} = -h(ma) + \delta \omega e^{-ma} \sum_{j: c < p_j} \eta_j h(\Lambda_j) \leq -h(ma) + e^{-ma} \leq 0.$$

The final inequality uses $h(x) \geq e^{-x}$, equivalent to $e^x - 1 \geq x$. Thus the type payoff difference is globally single crossing. Let $R(a) = D_r(a; a)$. At $a = p_1$, immediate surplus is zero and waiting surplus is nonnegative, so $R(p_1) \leq 0$. If $R(0) \leq 0$, $a = 0$ is a cutoff best response. Otherwise continuity and the intermediate value theorem give a root in $(0, p_1]$, and type single crossing supplies the full best-response inequalities.

Step 2: derivative bounds. On the active domain $p_1 < \beta$, total rider continuation mass is

$$\ell = \frac{1 - p_1/\beta}{1 - p_1} \leq 1.$$

On a strict repeat-rescue branch,

$$\eta_{2,a} = \frac{\omega m(p_2 - p_1)}{\beta(1 - p_1)(C_2 - C_1)} \geq 0, \quad \eta_1 = \ell - \eta_2.$$

Define

$$\mathcal{A}(a) = h(ma)(p_1 - a), \quad Y_j(a) = e^{-ma} h(\Lambda_j(a))(p_j - a).$$

The integral representation $h(z) = \int_0^1 e^{-zt} dt$ gives $h' < 0$, $h + h' > 0$, and $h(z) + zh'(z) = e^{-z}$. Put $y_j = \omega m(p_j - a) \leq \Lambda_j$. Then

$$\begin{aligned} -\omega e^{ma} Y_j' &= \omega h(\Lambda_j) + y_j[h(\Lambda_j) + \omega h'(\Lambda_j)] \\ &\leq \omega h(\Lambda_j) + \Lambda_j[h(\Lambda_j) + \omega h'(\Lambda_j)] \\ &= 1 - (1 - \omega)e^{-\Lambda_j} \leq 1. \end{aligned}$$

Hence

$$-\omega Y_j'(a) \leq e^{-ma}. \quad (7)$$

Also

$$-\mathcal{A}'(a) = h(ma) - m(p_1 - a)h'(ma) > e^{-ma}. \quad (8)$$

Step 3: every interior root is a strict downward crossing. At any nondegenerate interior residual root on an active rescue branch,

$$h(\Lambda_2)(p_2 - a) > h(\Lambda_1)(p_1 - a).$$

Otherwise discounted waiting is at most $\delta\omega e^{-ma}\ell(p_1 - a)$, strictly below $h(ma)(p_1 - a)$, contradicting indifference. Therefore $Y_2 > Y_1$ at a root. Let

$$\mathcal{T}(a) = (\ell - \eta_2)Y_1 + \eta_2 Y_2.$$

The composition derivative $\eta_{2,a}(Y_2 - Y_1)$ is nonnegative. Applying (7) gives

$$-\omega \mathcal{T}'(a) \leq \ell e^{-ma}.$$

Since $R = \mathcal{A} - \delta\omega \mathcal{T}$, equation (8) yields at every interior root

$$R'(a) < -e^{-ma} + \delta\ell e^{-ma} \leq 0.$$

Before rescue activates, only repeat at p_1 is relevant and, for every $a < p_1$,

$$R(a) = (p_1 - a) [h(ma) - \delta\omega\ell e^{-ma}h(\Lambda_1)] > 0.$$

Thus the rescue-activation point cannot itself be a root. Starting from the positive inactive branch, a continuous scalar residual whose every active root crosses strictly downward can have at most one root. Existence therefore implies uniqueness.

The boundary cases follow directly: if $p_1 = 0$, the cutoff domain is the singleton $a = 0$; if $p_1 \geq \beta$ or $\omega = 0$, waiting has no positive value and $a = p_1$; if $p_2 = p_1$, delay and weakly worse assignment dominate waiting, again giving $a = p_1$. The point $p_1 = 1$ is the continuous zero-demand closure.

For continuity, take any convergent sequence of policies and primitives and their unique cutoffs. Every subsequential cutoff limit satisfies the closed best-response inequalities of the limiting problem. Uniqueness identifies that limit, so the full sequence converges. Lemma 2 then gives continuity of completion. Under the deterministic closure, Q^{com} is continuous as well.

B Audited claim ledger

Claim	Status after adversarial audit
Conditional Poisson supply after universal rejection	Proved under marked Poisson costs and type-independent retention.
Unique full WPBE in every action label	Too strong. Unique incumbent cutoff and completion; payoff-equivalent labels may differ.
Exact rider completion formula	Proved globally with $\max\{0, A_1, A_2\}$.
Outer optimum exists	Proved for compact policies and the continuous committed-reach objective.
Both incremental gaps vanish in thin and thick limits	Proved after full WPBE response and reoptimization.
Any positive gap has an interior maximum	Proved; uniqueness is not implied.
Expanded-search value always peaks first	False as a universal fixed-menu claim; maintained-primitives numerical reversal exists.
Frozen-hazard Poisson peaks are ordered	Proved with closed forms.
Uniform separation preserves the order	Proved as a robustness certificate, not a primitive condition.
Finite no-search threshold under committed reach	Proved under uniform Lipschitz regularity.
Finite no-search threshold under executed contacts	False at a qualifying rescue-activation boundary.
$s^*(m)$ is globally monotone	False in numerical examples; an off-on-off pattern can arise.
The current objective is profit or welfare	False. It is completion value net of a search-resource cost.
$I \subset A \subset E$ as mechanism sets	False. Only $A \subset E$ through $s = 1$; I and A are supply environments.

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